

Ziyao (Adonis) Zeng | 曾子尧

Seeking research roles from Dec. 2026 | Last Update: Sept. 2026

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Research Interests

- **Multimodal Learning** inspired by humans for **Agents and Robotics**, especially **VLMs, LLM Agents, Diffusion (World) Models, 3D Vision**
- My line of work in “**Language for 3D Vision**” explores **how vision-language models can perceive and understand the world like humans do.**

Work Experience

- **Meta Reality Labs** May 2026 – Dec. 2026
 - Research Scientist Intern | Working with **Dr. Ziyi Kou** and Dr. Li Guan | **Dexterous Tactile VLMs** (CVPR 2027 Submission)
 - Built a video–tactile benchmark exposing VLMs’ blindness to touch-only properties
 - Proposed a training-free Tactile-as-Prompt method that beats learned encoders and scales to visual-tactile in-the-wild data
- **NVIDIA Research** May 2025 – March 2026
 - Research Intern | Worked with **Dr. Hang Su**, Dr. Jindong Jiang, Dr. Abhishek Badki, Dr. Orazio Gallo, and Dr. Sifei Liu | **Spatial VLMs**
 - Taught VLMs to implicitly reconstruct 3D scenes as latent chain-of-thought before answering spatial questions
 - Inventor on **U.S. patent** application “Boosting Spatial Understanding in VLMs with Spatial State Tokens” (Pending, NVIDIA Corp., 2026)
- **Shanghai AI Lab** Oct. 2022 – March 2023
 - Intern Researcher | Visual grounding and prototype-based segmentation
- **UISEE Technologies** July 2021 – April 2022
 - Intern Algorithm Engineer | Developed point cloud recognition & monocular 3D object detection algorithm for autonomous driving

Education

- **Yale University** Aug. 2023 – **[Expected] Dec. 2026**
 - **Ph.D. in Computer Science**, advised by **Prof. Yuval Kluger** and **Prof. Daniel Rakita**
- **ShanghaiTech University** Sept. 2019 – June 2023
 - **B.Eng. in Computer Science**, Major | Innovation and Entrepreneurship, Minor
 - Research Assistant at PLUS Lab, ShanghaiTech University; working with **Prof. Xuming He**
 - Research Assistant at **GRASP Lab, University of Pennsylvania**; working with **Prof. Jianbo Shi** and **Dr. Renrui Zhang**
 - Entrepreneurial Mindshift Program, Babson College, March 2021 | International Summer Program, Osaka University, June 2022

Selected Publications *(order by significance; *: equal contribution)*

- **Dexterous Tactile World Model** [World Models]
Ziyao Zeng*, Xiaotao Sun, Hao Wang, Yueyang Pan, Zhengxiang Yu, Fengyu Yang, Tianyu Liu, Zhiwen Fan, Daniel Rakita
Under Review | Link: <https://adonis-galaxy.github.io/dtwm-project-page/>
 - Whole-hand touch can improve future-frame predications of world models since touch history correlates with future manipulation.
- **3DCodeVerse: Exploring 3D Worlds as Code** [VLMs, LLM Agents, 3D Vision]
Yipeng Gao*, Ziyao Zeng*, Jinfa Huang*, Yijiang Li*, <14 more authors>, Laurent Itti
Under Review | Link: <https://3dcodeverse.com/>
 - An ecosystem for 3D generation via code including a 2M-program dataset and a multi-agent harness with automated spatial and visual verification.
- **RuleSmith: Multi-Agent LLMs for Automated Game Balancing** [LLM Agents]
Ziyao Zeng, Hao Wang, <8 more authors>, Daniel Rakita, Ronald R. Coifman, Yuval Kluger, Zhiwen Fan
NeurIPS 2026 | NE Agents Day 2026 | Link: <https://adonis-galaxy.github.io/RuleSmith-website/>
 - Multi-agent LLMs self-play with Bayesian Optimization to automatically balance numerical parameters in asymmetric strategic games.
- **SciAgentArena: Benchmarking AI Agents for Addressing Scientific Challenges Across Scales** [LLM Agents]
Tianyu Liu*, <6 more co-first authors>, Ziyao Zeng*, <21 more authors>, Marinka Zitnik, James Zou, Hua Xu, Hongyu Zhao
Under Review of Nature | Link: <https://arxiv.org/abs/2606.12736>
 - We introduce **SciAgentArena**, a benchmark with ~200 tasks for evaluating AI agents across diverse **real-world scientific research scenarios**.
- **Iris: Integrating Language into Diffusion-based Monocular Depth Estimation** [VLMs, Diffusion Models, 3D Vision]
Ziyao Zeng*, Jingcheng Ni*, Daniel Wang, Patrick Rim, Younjoon Chung, Fengyu Yang, Byung-Woo Hong, Alex Wong
CVPR 2026 | NECV 2025 Oral Presentation (18.75%) | Link: <https://adonis-galaxy.github.io/Iris-website/>
 - We find **language improves diffusion-based depth estimators** through the conditional distributions modeled in text-to-image pre-training.
- **RSA: Resolving Scale Ambiguities in Monocular Depth Estimators through Language Descriptions** [VLMs, 3D Vision]
Ziyao Zeng, Yangchao Wu, Hyoungseob Park, Daniel Wang, Fengyu Yang, Stefano Soatto, Dong Lao, Byung-Woo Hong, Alex Wong
NeurIPS 2024 | Link: <https://arxiv.org/abs/2410.02924>
 - The **first** to use language to infer scales of 3D scenes, enabling **zero-shot relative-to-metric depth alignment**.
- **WorDepth: Variational Language Prior for Monocular Depth Estimation** [VLMs, 3D Vision]
Ziyao Zeng, Daniel Wang, Fengyu Yang, Hyoungseob Park, Yangchao Wu, Stefano Soatto, Byung-Woo Hong, Dong Lao, Alex Wong
CVPR 2024 | Link: <https://arxiv.org/abs/2404.03635>
 - We formulate **language as a variational prior** by text-VAE, then use **image as condition** to select most likely depth as **posterior**.
- **Can Language Understand Depth?** [VLMs, 3D Vision]
Renrui Zhang*, Ziyao Zeng*, Ziyu Guo, Yafeng Li
ACM Multimedia 2022 (Brave New Idea, 12.5%) | Link: <https://arxiv.org/abs/2207.01077>
 - The **first** to conduct **zero-shot training-free adaptation** from **VLMs to monocular depth estimation**.

Other Publications *(selected; order by significance; *: equal contribution)*

- **Coffee: Controllable Diffusion Fine-tuning** [Diffusion Models]
Ziyao Zeng, Jingcheng Ni, Ruyi Liu, Alex Wong | Under Review | Link: <https://arxiv.org/abs/2511.14113>
- Coffee controls diffusion fine-tuning to **prevent learning and entangling undesired concepts**.
- **OpenLongTail: Generative Scaling of Long-Tail Driving Data** [World Models]
Lulin Liu, <7 more authors>, Ziyao Zeng, <7 more authors>, Marco Pavone, Zhiwen Fan | Under Review | Link: <https://arxiv.org/abs/2607.09655>
- A world model that scales **monocular in-the-wild long-tail driving videos into multi-view assets** for long-tail training.
- **UniTac: A Unified Multimodal Model for Cross-Sensor Tactile Understanding and Generation** [VLMs, Multimodal]
Jiahang Tu, Fengyu Yang, Chenyang Ma, Xihang Yu, Ziyao Zeng, Shaokai Wu, Hanbin Zhao, Zhi Tao, Chao Zhang, Hui Qian, Alex Wong
ECCV 2026 (Oral) | Link: <https://arxiv.org/abs/2606.31451>
- The **first unified multimodal model** designed for **tactile understanding and generation**.
- **A Physics-Grounded Benchmark for Multi-Agent Dynamics in World Models** [World Models]
Nuo Chen, Lulin Liu, Zihao Li, Ziyao Zeng, <6 more authors>, Boris Ivanovic, Marco Pavone, Zhangyang Wang, Yang Zhou, Zhiwen Fan
ECCV 2026 (Oral) | Link: <https://arxiv.org/abs/2606.28757>
- Introduce **CrashTwin**, a **physics-grounded** world model evaluation framework with **25K synthetic** and **12K in-the-wild** collision sequences.
- **Artificial Foveated Perception for Mitigating Shortcut Learning in Robotic Foundation Models** [VLMs, Robotics]
Xiatao Sun, <6 more authors>, Ziyao Zeng, Shawn Li, Qian Wang, Fei Miao, Daniel Rakita
CoRL 2026 | Link: <https://arxiv.org/abs/2607.10655>
- Integrates Artificial Foveated Perception into VLA and World Action Model can **suppress spurious scene-level correlations** to improve generalization.
- **4DP-QA: Scalable QA for 4D Perception in Vision Language Models** [VLMs, 3D Vision]
Seokju Cho, Abhishek Badki, Hang Su, Jindong Jiang, Ziyao Zeng, Seungryong Kim, Sifei Liu, Orazio Gallo
CVPR 2026 | Link: <https://arxiv.org/abs/2606.11568>
- Propose **True-Motion Tracking for VLM** for **motion-related scene understanding**; construct **400K training data** and a **2.2K-sample benchmark**.
- **ETA: Energy-based Test-time Adaptation for Depth Completion** [3D Vision]
Younjoon Chung, <4 more authors>, Ziyao Zeng, Safa Cicek, Byung-Woo Hong, James S. Duncan, Alex Wong
ICCV 2025 | Link: <https://arxiv.org/abs/2508.05989>
- Energy-based test-time adaptation of pretrained depth completion models by quantifying the likelihood.
- **ProtoDepth: Unsupervised Continual Depth Completion with Prototypes** [3D Vision]
Patrick Rim, Hyoungseob Park, S. Gangopadhyay, Ziyao Zeng, Younjoon Chung, Alex Wong
CVPR 2025 | Link: <https://arxiv.org/abs/2503.12745>
- A prototype-based approach that learns domain descriptors for continual learning for unsupervised depth completion
- **Binding Touch to Everything: Learning Unified Multimodal Tactile Representations** [VLMs, Multimodal]
Fengyu Yang*, Chao Feng*, Ziyang Chen*, <3 more authors>, Ziyao Zeng, Xien Chen, Rit Gangopadhyay, Andrew Owens, Alex Wong
CVPR 2024 | Link: <https://cfeng16.github.io/UniTouch/>
- Learn a **unified multimodal tactile representation**, enabling touch-LLM, touch-to-image generation, image texture manipulation.
- **PointCLIP V2: Adapting CLIP for Powerful 3D Open-world Learning** [VLMs, 3D Vision]
Xiangyang Zhu*, Renrui Zhang*, Bowei He, Ziyao Zeng, Shanghang Zhang, Peng Gao
ICCV 2023 | Link: <https://arxiv.org/abs/2211.11682>
- Adapt CLIP for open-world 3D understanding and exceed PointCLIP by up to **42.90% accuracy**; using **realistic shape projection** and LLMs.
- **iQuery: Instruments as Queries for Audio-Visual Sound Separation** [Multimodal]
Jiabao Chen, Renrui Zhang, Dongze Lian, Jiaqi Yang, Ziyao Zeng, Jianbo Shi
CVPR 2023 | Link: <https://arxiv.org/abs/2212.03814>
- Propose Instrument as Query (iQuery) with a flexible query expansion mechanism, achieving up to 44.2 % improvement.

Service / Teaching / Awards

Reviewer: CVPR, ICCV, ECCV, ICML, ICLR, NeurIPS, ACM MM, AISTATS, ICASSP, BMVC, ICRA, TCSVT, TMLR;
Outstanding Reviewer of CVPR 2025
Peer Mentor, Yale CS PhD Class of 2024; Intern Lecturer at Shanghai Advanced Research Institute, 2021
Teaching Assistant for Introduction to Machine Learning & Introduction to AI Application, Yale University
Sigma Xi, The Scientific Research Honor Society, Membership Nomination

2025/2026
2025

Skills

Frameworks & languages: Python/PyTorch (distributed training, diffusion/DiT fine-tuning, VLM post-training for spatial & motion reasoning), LLM API integration (tool calling, structured output), agentic coding (Claude Code), C/C++, Unity/C#
Model families: Qwen, InternVL, NVILA, CLIP, Wan, Stable Diffusion
Generative & adaptive modeling: diffusion, VAE, energy-based models, world models; test-time adaptation, continual & unsupervised learning; Bayesian optimization; multi-agent LLM systems
3D & multimodal: monocular depth, point clouds, multi-view geometry; tactile, audio & neural signals
Data & benchmarks: driving simulation & synthetic data; build large-scale benchmark (200-task agent benchmark, 400K VQA)